

Cassie Charlie

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SUMMARY

Computer Science graduate with past research experience in AI/ML and data analytics, including an NSTC grant for speech synthesis. Proficient in Python, PyTorch, SQL, and Power BI. Seeking entry-level roles in data analytics or AI engineering.

EDUCATION

National Dong Hwa University (國立東華大學)	Hualien, Taiwan
<i>Bachelor of Computer Science and Information Engineering</i>	<i>Graduation Date: Jun 2026</i>

WORK EXPERIENCE

National Dong Hwa University	Hualien, Taiwan
<i>Teaching Assistant Introductory Machine Learning</i>	<i>Oct 2025 - Jan 2026</i>

- Managed classroom logistics for 45 students including equipment setup, attendance tracking, and hands-on ML session support.
- Developed homework solutions and grading rubrics; evaluated 150+ assignments and exams with consistent feedback and fair assessment.
- Conducted office hours and review sessions, helping students debug Python code and understand core ML concepts.

National Dong Hwa University	Hualien, Taiwan
<i>Research Assistant at Prof. Wen-Chieh Fang's Lab</i>	<i>Jun 2025 - Nov 2025</i>

- Conducted ablation studies and hyperparameter optimization for diffusion-based speech enhancement, achieving 38.6% CSIG improvement and 16.6% STOI gain over baseline configurations.
- Benchmarked architectural variants (DiffuSE vs DiffuSEC) across 6 evaluation metrics (PESQ, CSIG, CBAK, COVL, SSNR, STOI), identifying optimal conditioning strategies for zero-shot audio deepfake enhancement.
- Developed chunked inference pipeline with overlap blending, enabling 94% ASV pass rate while maintaining audio quality for full-length samples.
- Co-authored research paper targeting journal/conference publication; responsible for experimental design, metric implementation, and results visualization.

PROJECT EXPERIENCE

National Science and Technology Council	Hualien, Taiwan
<i>EmoScale NSTC-Funded Undergraduate Research</i>	<i>Jul 2025 - Mar 2026</i>

- Designed intensity learning framework using RankNet on prosodic features (OpenSMILE), achieving 92.45% validation accuracy in predicting relative emotional intensity without manual labels.
- Architected embedding extraction pipeline (HuBERT content, ECAPA-TDNN speaker, Wav2Vec2 emotion) supporting diffusion decoder training for 6 speakers with zero-shot evaluation on 2 unseen speakers.
- Validated intensity controllability via RMSE analysis across scales 0.0-1.0 (increase from 0.000 to 0.0137), confirming model's responsiveness to conditioning signals.

SKILLS & LANGUAGES

Technical Skills	- Programming Languages: Python (Numpy, Pandas), C, C++ - Data Analytics: SQL, Power BI, Excel - Machine Learning/AI: PyTorch, Diffusion Models, Speech Processing (ASR/TTS) - Tools: Linux, Git, VS Code
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Languages: Indonesian (Native), English (Fluent, IELTS 8.0), Chinese (Intermediate)